miRNA profiles in early systemic sclerosis, organ complications, and treatment response

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To the Editor: Systemic sclerosis (SSc) is a complex autoimmune disease characterized by immune dysfunction and excessive collagen deposition, leading to fibrosis across multiple organs and tissues.[1] Key features include skin thickening, Raynaud's phenomenon (RP), and internal organ involvement, with interstitial lung disease (ILD), pulmonary arterial hypertension, renal crisis, and myocardial involvement. Early diagnosis of SSc complications is crucial for effective management and improved patient outcomes, emphasizing the necessity of research to identify robust diagnostic indicators.

MicroRNAs (miRNAs), a class of short noncoding RNA molecules, have emerged as key players in the pathogenesis of SSc. Specific miRNAs signatures exhibit the potential to serve as diagnostic and prognostic biomarkers. However, existing studies on serum miRNA expression in SSc are limited by the number of detected miRNAs, size of the SSc patient cohort, and a lack of information on miRNA profiles after treatment. In this study, we conducted miRNA sequencing on serum samples obtained from patients with SSc both before and after a 6-month treatment period and matched healthy controls (HCs). We also investigated the clinical relevance of dysregulated miRNAs and validated differentially expressed (DE) miRNAs in an independent, large cohort.

This study was approved by the Medical Ethics Committee of Central South University Xiangya Hospital (201303293). Informed consent was obtained from all participants.

Detailed information on study population, sample collection and storage, serum miRNA library construction, sequencing, data processing, bioinformatic analysis of the miRNA sequencing data, and real-time quantitative polymerase chain reaction, as well as statistical analysis, is presented in Supplementary Methods, http://links.lww.com/CM9/C389.

Seventeen treatment-naïve patients with SSc and 15 age- and sex-matched HCs were included for miRNA sequencing [Supplementary Table 1, http://links.lww.com/CM9/C389]. A total of 111 DE serum miRNAs were identified between SSc patients and HCs [Supplementary Figure 1 and Supplementary Table 2, http://links.lww.com/CM9/C389]. Notably, miR-99b-5p, miR-30a-5p, miR-1-3p, miR-149-5p, miR-187-3p and let-7c-5p were significantly positively correlated with SSc [Figure 1A]. Gene Ontology (GO) analysis revealed significant enrichment in the Wnt and transforming growth factor-β (TGF-β) signaling pathways, with specific DE miRNAs, including miR-1-3p, miR-203a-3p, miR-30a-5p, miR-372-3p, and miR-149-5p, involved in both pathways [Supplementary Figure 1, http://links.lww.com/CM9/C389].

To explore the roles of miRNAs in SSc-related organ complications, we compared miRNA sequencing data between patients with and without RP and ILD [Figure 1B and Supplementary Figure 2, http://links.lww.com/CM9/C389]. In patients with RP, miR-2115-5p is upregulated and positively associated with RP. In contrast, miR-1-3p, miR-133a-3p, miR-615-3p, miR-708-3p, miR-203a-3p, and let-7c-5p were downregulated and negatively correlated with RP. GO functional analysis of the top miRNAs (targeting 689 genes) identified pathways related to endothelial cell migration, proliferation, and vasculogenesis. Similarly, several miRNAs were positively associated with

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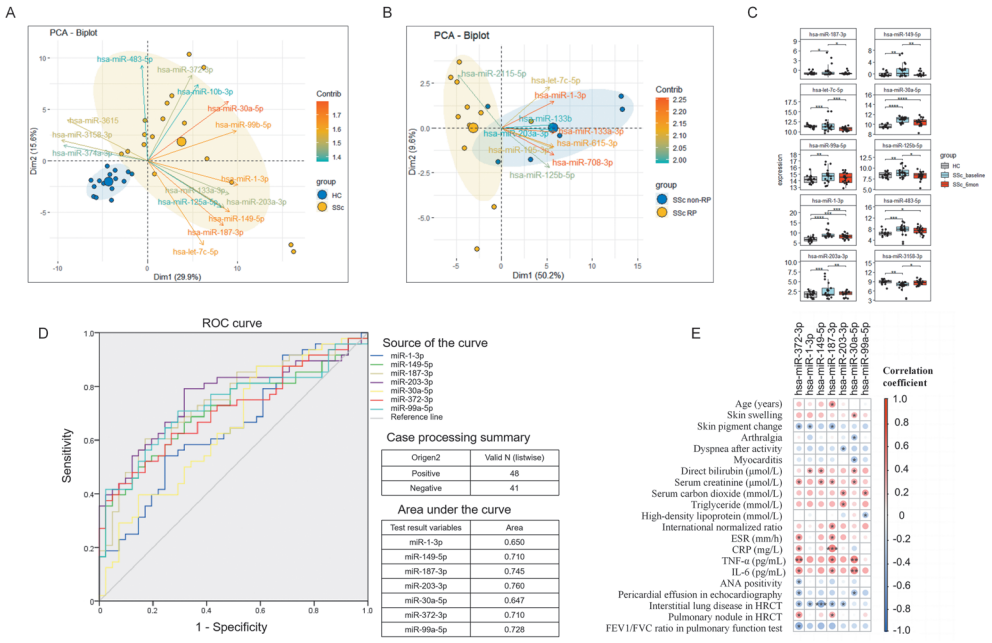

Origen2	Valid N (listwise)
Positive	48
Negative	41

Test result variables	Area
miR-1-3p	0.650
miR-149-5p	0.710
miR-187-3p	0.745
miR-203-3p	0.760
miR-30a-5p	0.647
miR-372-3p	0.710
miR-99a-5p	0.728

Figure 1: Serum DE miRNA profiles in SSc and the correlation analysis with clinical characteristics. (A) The PCA biplot indicated the separation of NCs and SSc by DE miRNAs. The blue and the yellow ellipses represented principal components of HCs and SSc patients, respectively. The direction of each arrow indicated the relationship between the miRNA variables and the principal components. Specifically, an miRNA variable was considered positively correlated with a principal component if its arrow pointed toward that principal component. And the closer the arrow's color is to red, the greater the contribution of this miRNA variable to predicting the corresponding principal component. (B) The PCA biplots indicated the separation of SSc patients by RP. The yellow and the blue ellipses represented principal components of SSc patients with and without RP, respectively. (C) The box plot showed the difference in expression of DE miRNAs among NCs and SSc patients before and after treatment. (D) The curves for these DE miRNAs. The values of these ROC curves ranged from 0.647 to 0.760. Notably, the closer the AUC value is to 1.0, the stronger the miRNA's ability to differentiate between SSc patients and HCs. (E) Correlation analysis of the clinical features/laboratory results/examination results and serum levels of selected seven miRNAs, performed by the Spearman correlation analysis. ANA: Anti-nuclear antibody; AUC: Area under the curve; CRP: C-reactive protein; DE: Differentially expressed; ESR: Erythrocyte sedimentation rate; FEV1/FVC: Forced expiratory volume in 1 second/forced vital capacity; HC: Healthy control; HRCT: High-resolution computed tomography; IL-6: Interleukin-6; miRNA: Micro RNA; PCA: Principal component analysis; ROC: Receiver operating characteristic; RP: Raynaud's phenomenon; SSc: Systemic sclerosis; SSc_6mon: SSc patients after 6-month treatment; SSc_baseline: SSc patients at baseline; TNF-α: Tumor necrosis factor-α. $^{*}P<0.05$; $^{\dagger}P<0.01$; $^{\ddagger}P<0.001$; $^{\S}P<0.0001$.

ILD, such as miR-1-3p, miR-708-3p, miR-149-5p, miR-203a-3p, miR-218-5p, miR-187-3p, and let-7c-5p.

Following treatment, a significant reduction in the modified Rodnan skin score was observed in 17 patients with SSc, indicating improvement in most cases. To identify DE miRNAs with expression patterns reversed post-treatment, we compared miRNA profiles among HCs and patients with SSc at baseline and after 6 months of treatment [Figure 1C and Supplementary Figure 3, http://links.lww.com/CM9/C389]. Notably, miR-187-3p, miR-149-5p, miR-30a-5p, miR-99a-5p, miR-1-3p, miR-203a-3p, let-7c-5p, miR-483-5p, and miR-125b-5p were upregulated at baseline but were downregulated after treatment, whereas miR-3158-3p showed the opposite trend.

Furthermore, we confirmed the DE miRNAs in the serum samples of another SSc cohort of 47 treatment-naïve patients (39 with serum samples both at baseline and after treatment) and 41 age- and sex-matched HCs using quantitative polymerase chain reaction (qPCR), and the characteristics of included participants are presented in Supplementary Table 3, http://links.lww.com/CM9/C389. Serum levels of miR-187-3p, miR-372-3p, miR-149-5p, miR-30a-5p, miR-99a-5p, miR-1-3p, and miR-203a-3p were upregulated in patients with SSc at baseline but reduced following treatment and have the potential to effectively differentiate SSc patients from HC [Supplementary Figure 4 and Figure 1D, http://links.lww.com/CM9/C389].

We observed significant associations between serum levels of the seven DE miRNAs and various clinical characteristics [Figure 1E]. Serum levels of miR-372-3p, miR-187-3p, and miR-30a-5p were positively correlated with inflammatory biomarkers in patients with SSc, including erythrocyte sedimentation rate, C-reactive protein, tumor necrosis factor-α, and interleukin-6. In addition, cutaneous swelling and edema, indicative of skin inflammation, were positively correlated with serum levels of miRNA-30a-5p. Conversely, abnormal skin pigmentation, dyspnea after activity, ILD, and cardiac manifestations reflecting tissue fibrosis were negatively correlated with serum levels of miR-372-3p, miR-187-3p, miR-149-5p, miR-30a-5p, miR-1-3p, and miR-203a-3p in patients with SSc. Renal involvement, indicated by elevated serum creatinine levels, positively correlated with miR-372-3p, miR-187-3p, miR-149-5p, and miR-30a-5p levels.

Besides, we summarized eight studies focusing on circulating cell-free DE miRNA profiles in patients with SSc [Supplementary Table 4, http://links.lww.com/CM9/C389], and found a total of 46 upregulated and 81 downregulated miRNAs. Among these DE miRNAs, miR-132, miR-145, and miR-155 were upregulated in patients with SSc in two studies; miR-191, miR-142-3p, miR-145-5p, miR-26a, miR-30b, miR-409-3p, miR-410, ENSG00000252277, and let-7d were downregulated in two studies; and miR-181a was downregulated in three

studies. However, discrepancies remain, as some studies reported contradictory expression patterns of miR-143, miR-423-5p, miR-223, miR-320b, miR-181b, and miR-29b-3p in patients with SSc.

In this study, we identified 111 DE serum miRNAs in treatment-naïve patients with SSc compared with those in HCs, among which 11 miRNAs have been reported in previous studies. In both our and previous studies, the expression of miR-483-5p, a potential profibrotic miRNA, was significantly elevated in patients with SSc. Previous studies have reported that miR-483-5p overexpression in fibroblasts and endothelial cells modulates the expression of fibrosis-related genes (*COL4A1*, *COL4A2*, *COL2A1*, *α-SMA*, and *SM22A*).[2] In addition, miR-409-3p and miR-4508 were significantly downregulated in our study [Supplementary Table 2, http://links.lww.com/CM9/C389] and in prior literature, especially miR-409-3p, which was reported in two other studies. However, the role of these miRNAs in SSc remains unclear. Notably, eight of the 11 miRNAs were upregulated in our study but downregulated in previous studies. Furthermore, we identified RP- (miR-2115-5p, miR-1-3p, miR-203a-3p, and let-7c-5p) and ILD-related miRNAs (miR-1-3p, miR-149-5p, miR-203a-3p, miR-187-3p, miR-125a-5p, and let-7c-5p) in SSc. Consistent with previous studies, serum miR-203a-3p was downregulated in SSc-associated RP.[3,4] However, miR-149-5p has been reported to be downregulated in SSc-associated ILD in previous study,[4] which contradicts the results of our study.

Inconsistencies in the expression of specific miRNAs exist between different studies. This may be attributed to small sample sizes and selection biases. SSc is a rare disease with an incidence rate of 125 per one million population. This study has strict inclusion and exclusion criteria to ensure the reliability of the results. And, none of the recruited patients had been previously treated with immunosuppressants. In addition, all candidates had a disease duration of no more than two years or evidence of disease progression (e.g., worsening skin stiffness or pulmonary fibrosis) over the past two years. Besides, sample size included in this study exceeded most studies and was statistically adequate.

In the correlation study, we found that miR-372-3p, miR-187-3p, and miR-30a-5p were probably related to inflammatory states, whereas miR-372-3p, miR-187-3p, miR-149-5p, miR-30a-5p, miR-1-3p, and miR-203a-3p were negative biomarkers of tissue fibrosis. In addition, miR-372-3p, miR-187-3p, miR-149-5p, and miR-30a-5p were associated with acute renal involvement in SSc. This suggests that these miRNAs participate in SSc pathogenesis. Some studies have reported the antifibrotic effects of miR-372-3p, miR-30a-5p, miR-149-5p, miR-1-3p, miR-203-3p, and miR-99a-5p in other diseases. Silencing miR-30a-5p ameliorates the degree of cardiac fibrosis in a viral myocarditis model.[5] In addition, miR-372-3p has been implicated in vasculopathy and inflammation mechanisms in other diseases. Specifically, miR-372-3p modulates high glucose-induced glomerular endothelial dysfunction in diabetic nephropathy.[6] Therefore, the roles of these miRNAs in tissue fibrosis and endothelial dysfunction may provide future treatment targets for SSc. Further research is required to elucidate these underlying mechanisms.

In this study, we compared serum DE miRNA profiles before and after treatment. The expression levels of seven miRNAs (miR-187-3p, miR-149-5p, miR-30a-5p, miR-99a-5p, miR-1-3p, miR-203a-3p, and miR-372-3p) were further validated by qPCR, which were upregulated in patients with SSc at baseline but reduced following treatment. This suggests that these miRNAs are promising biomarkers for treatment response monitoring.

In conclusion, the expression levels of miR-187-3p, miR-149-5p, miR-30a-5p, miR-99a-5p, miR-1-3p, miR-203a-3p, and miR-372-3p in serum may serve as promising markers for early diagnosis and treatment response monitoring in patients with SSc. Further investigation of these miRNAs may enhance our understanding of the underlying mechanisms of SSc.

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Conflicts of interest

None.

References

1. Zhu H, Luo H, Skaug B, Tabib T, Li YN, Tao Y, *et al*. Fibroblast subpopulations in systemic sclerosis: Functional implications of individual subpopulations and correlations with clinical features. J Invest Dermatol 2024;144:1251–1261.e13. doi: 10.1016/j.jid.2023.09.288.
2. Chouri E, Servaas NH, Bekker C, Affandi AJ, Cossu M, Hillen MR, *et al*. Serum microRNA screening and functional studies reveal miR-483-5p as a potential driver of fibrosis in systemic sclerosis. J Autoimmun 2018;89:162–170. doi: 10.1016/j.jaut.2017.12.015.
3. Kanno Y, Shu E, Niwa H, Seishima M, Ozaki KI. MicroRNA-30c attenuates fibrosis progression and vascular dysfunction in systemic sclerosis model mice. Mol Biol Rep 2021;48:3431–3437. doi: 10.1007/s11033-021-06368-z.
4. He Y, Liu H, Wang S, Chen Y. In silico detection and characterization of microRNAs and their target genes in microRNA microarray datasets from patients with systemic sclerosis-interstitial lung disease. DNA Cell Biol 2019;38:933–944. doi: 10.1089/dna.2019.4780.
5. Zhang Y, Cai S, Ding X, Lu C, Wu R, Wu H, *et al*. MicroRNA-30a-5p silencing polarizes macrophages toward M2 phenotype to alleviate cardiac injury following viral myocarditis by targeting SOCS1. Am J Physiol Heart Circ Physiol 2021;320:H1348–H1360. doi: 10.1152/ajpheart.00431.2020.
6. Meng Z, Li F, Wang B. miR-372-3p is a potential diagnostic factor for diabetic nephropathy and modulates high glucose-induced glomerular endothelial cell dysfunction via targeting fibroblast growth factor-16. Arch Med Sci 2023;19:703–716. doi: 10.5114/aoms.2019.89659.

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